

Analysis PhD Exam, September 1991

Present all work in a neat and logical fashion.

1. State and prove the Lebesgue Monotone Convergence Theorem.
2. Prove that $L^1(S, \Sigma, \mu)$ is complete, where μ is a nonnegative measure.
3. Give the construction of the product measure $\mu \times \nu$ on $\mathcal{S} \times \mathcal{W}$, where $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{W}, \nu)$ are σ -finite nonnegative measure spaces.
4. Let f be Lebesgue integrable on $\mathbb{R}$. Suppose $\int_0^x f \, dm = 0$ for all real x . What can you say about f ? Prove your answer.
5. Let f be a Lebesgue integrable function on $\mathbb{R}$. Let $\epsilon > 0$. Is it possible to find an interval I such that whenever E is a measurable set such that $E \cap I = \emptyset$, then $|\int_E f \, dm| < \epsilon$?
6. Let X be a metric space. Let $\{P_n\}$ be a sequence of regular probability measures on $\mathcal{B}(X)$, the Borel subsets of X . Let $P = \sum_{n=1}^{\infty} (\frac{1}{2^n}) P_n$. Show that P is regular on $\mathcal{B}(X)$.
7. Let $\mu(dx) = x^2 dx$ on $[0, 1]$. Let $T : [0, 1] \rightarrow [0, 1]$ be defined by $T(x) = x^4$. Compute the Radon–Nikodym derivative of the image measure $T\mu$.
8. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space with $\mu(\Omega) = 1$. Let $\mathcal{A}$ and $\mathcal{B}$ be sub σ -algebras of $\mathcal{F}$. Suppose $\mu(A \cap B) = \mu(A)\mu(B)$ for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$. Prove that $\int fg \, d\mu = (\int f \, d\mu)(\int g \, d\mu)$ for all $\mathcal{A}$ -measurable positive functions f and all $\mathcal{B}$ -measurable positive functions g .